Supplementary material: Local Birkhoff structure and full-coil control of a three-dimensional magnetic rod

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Stage 3–5 evidence; repository base 1c73a58

This supplement expands the existing local proofs and evidence boundaries used in the article. It contains no Stage 6 or later bridge result. The symbols and equations are identical in the Chinese companion. Source comments and the evidence manifests identify the pre-existing proofs and computations; no new scientific experiment was run for this manuscript.

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1 Complete local proofs for transported structure and generating maps

This supplement supplies the local proofs underlying the theory section. Throughout, $z = (Q, P) \in U \subset \mathbb{R}^{2n}$, $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, $K_0 = -J$, and $(Da)_{ij} = \partial_{z_j} a_i$. The matrix M denotes normalized mass. Derivatives are taken at fixed physical time unless indicated otherwise. A time subscript on a differential form denotes its time section; ∂_τ denotes differentiation. Domains are restricted as necessary to make flows, inverse maps, and implicit branches single-valued. These are local results, with the regularity and uniform estimates stated explicitly below.

1.1 Variational signs, compatibility, and transported volume

Let a have two continuous spatial derivatives and one continuous time derivative, and let B be differentiable. For a variation $z + \epsilon\xi$ with ξ zero at the endpoints,

$$\left.\frac{d}{d\epsilon}\right|_0 \int \{a(z+\epsilon\xi,\tau)^T(\dot z+\epsilon\dot\xi) - B(z+\epsilon\xi,\tau)\}\,d\tau = \int \{\xi^T Da^T\dot z + a^T\dot\xi - \xi^T\nabla B\}\,d\tau \tag{S1}$$

$$= \int \xi^T\{(Da^T - Da)\dot z - \partial_\tau a - \nabla B\}\,d\tau. \tag{S2}$$

The fundamental lemma gives $K\dot z = \nabla B + \partial_\tau a$ with $K = Da^T - Da$. In particular the convention $D = -\nabla B - \partial_\tau a$ yields $\dot z = -K^{-1}D$, provided K is invertible. Independently,

$$d(a^T dz) = \sum_{i,j} \partial_{z_j} a_i\, dz_j \wedge dz_i = \frac{1}{2}\sum_{i,j}(\partial_{z_i}a_j - \partial_{z_j}a_i)\, dz_i \wedge dz_j. \tag{S3}$$

The factor $1/2$ is essential when both indices are summed. It gives $\Omega(v,w) = v^T K w$, and hence $\iota_f\Omega = -(Kf)^T dz$.

Since $d\Omega = 0$, Cartan's formula implies

$$\partial_\tau\Omega + \mathcal{L}_f\Omega = d((\partial_\tau a)^T dz) - d((Kf)^T dz) = -d(dB) = 0. \tag{S4}$$

In coordinates this is $K_\tau + DK[f] + Df^T K + KDf = 0$. Differentiating the flow pullback therefore gives $\frac{d}{d\tau}\Phi_{\tau,s}^*\Omega_\tau = 0$, proving $\Phi_{\tau,s}^*\Omega_\tau = \Omega_s$. Wedge powers give preservation of $\Omega_\tau^n/n!$. For a near-identity orientation-preserving flow, taking determinants in the matrix pullback yields

$$\det D\Phi_{\tau,s}(z)\,\frac{\sqrt{|\det K_\tau(\Phi_{\tau,s}z)|}}{\sqrt{|\det K_s(z)|}} = 1. \tag{S5}$$

This does not require the Euclidean determinant to equal one. For uniform damping of n momenta, it equals $e^{-n\gamma(\tau-s)}$, while the displayed density ratio equals $e^{n\gamma(\tau-s)}$.

1.2 Homogeneous transport, the sourced equation, and truncation

If f generates a smooth local invertible flow and $d\alpha_s$ is nondegenerate, define $\alpha_\tau = (\Phi_{\tau,s}^{-1})^*\alpha_s$. Pullback preserves closedness and nondegeneracy. Differentiation yields $\partial_\tau a + Daf + Df^T a = 0$. Taking $B = a^T f$ gives

$$\nabla B = Da^T f + Df^T a, \qquad \nabla B + \partial_\tau a = (Da^T - Da)f. \tag{S6}$$

Thus this is an existence proof of a Birkhoff representation. It is not an algorithm that presupposes access to the unknown flow.

Let $\theta = \tau - s$ and suppose $f = \sum_{k\ge0}\theta^k f_k$, $a = \sum_{j\ge0}\theta^j a_j$. The coefficient of θ^m in the transport equation is

$$(m+1)a_{m+1} + \sum_{j+k=m}(Da_j f_k + Df_k^T a_j) = 0. \tag{S7}$$

This proves the nonautonomous coefficient recursion, with factorials included in a_j, f_k. For an autonomous field, $a_{j+1} = -\mathcal{L}_f a_j/(j+1)$. Substitution of $a^{[r_o]} = \sum_{j=0}^{r_o}\theta^j a_j$ leaves only $\partial_\tau a^{[r_o]} + \mathcal{L}_f a^{[r_o]} = \theta^{r_o}\mathcal{L}_f a_{r_o}$. Its representation residual is the negative of this expression. Analyticity on a common complex neighborhood with suitable bounds is required for an infinite-series convergence claim. Finite differentiability suffices to compute finitely many coefficients and state a finite remainder.

For $h = T+V$, $T = P^T M(Q)^{-1}P/2$, and $f = (h_P, -h_Q + r)$, set $a_0 = (P, 0)$ and $\ell = P^T h_P - h = T - V$. A componentwise calculation gives

$$(Da_0 f)_Q = -h_Q + r, \qquad (Da_0 f)_P = 0, \tag{S8}$$

$$(Df^T a_0)_Q = (D_Q h_P)^T P, \qquad (Df^T a_0)_P = (D_P h_P)^T P. \tag{S9}$$

The gradient of $\ell = P^T h_P - h$ has precisely these conservative components, because the derivative of $P^T h_P$ in momentum contains an additional h_P that cancels the derivative of h. Therefore

$$\mathcal{L}_f a_0 = d\ell + (r, 0). \tag{S10}$$

The notation $d\ell$ here denotes its coefficient vector. Taking $a' = a_0 + b$ and requiring $\partial_\tau a' + \mathcal{L}_f a' = d\ell$ yields

$$b_0 = 0, \quad b_1 = -(r, 0), \quad b_{j+1} = -\mathcal{L}_f b_j/(j+1), \qquad B' = a'^T f - \ell = h + b^T f. \tag{S11}$$

For a truncation at r_o, all lower powers cancel and

$$e_B = K^{[r_o]} f - \partial_\tau a'^{[r_o]} - \nabla B'^{[r_o]} = -\theta^{r_o} \mathcal{L}_f b_{r_o}, \tag{S12}$$

$$f_a = (K^{[r_o]})^{-1}(\nabla B'^{[r_o]} + \partial_\tau a'^{[r_o]}) = f - (K^{[r_o]})^{-1} e_B. \tag{S13}$$

For $r = 0$, all $b_j = 0$, so $a' = a_0$ and $B' = h$ are already exact.

To verify the gauge relation, solve locally $\sigma_\tau + f^T \nabla\sigma = \ell, \sigma(s) = 0$. Exterior differentiation commutes with the Lie derivative, so $a = a' - d\sigma$ solves homogeneous transport. Moreover,

$$B = B' + \sigma_\tau = a'^T f - \ell + \sigma_\tau = (a' - d\sigma)^T f = a^T f. \tag{S14}$$

For any gauge $a \mapsto a + d\sigma$, $B \mapsto B - \sigma_\tau$, both K and e_B are unchanged by cancellation of mixed derivatives. This fact does not allow a gauge change to repair a nonzero residual. Indeed, with $\sigma_1 = \ell$, $\sigma_{j+1} = -(f^T \nabla\sigma_j)/(j+1)$ and $a_j = b_j - d\sigma_j$, a finite $\sigma^{[r_o]}$ obeys

$$\sigma_\tau^{[r_o]} + f^T \nabla\sigma^{[r_o]} = \ell + \theta^{r_o} f^T \nabla\sigma_{r_o}. \tag{S15}$$

Transforming the finite homogeneous pair with this gauge therefore gives the sourced scalar minus $\theta^{r_o} f^T \nabla\sigma_{r_o}$. The source construction truncates a different but infinitely gauge-equivalent equation; it is not a claim that a finite gauge transformation removes truncation error.

1.3 Spatial coefficient approximation and the represented tangent

The implementation approximates the coefficient functions, rather than the original vector field, by polynomials

$$\widetilde{b}_j(z) = b_j(z_*) + Db_j(z_*)(z - z_*) + \tfrac{1}{2} D^2 b_j(z_*)[z - z_*, z - z_*], \tag{S16}$$

with a zero quadratic term when only linear coefficients have been computed. Form $\widetilde{b} = \sum_{j=1}^{r_o} \theta^j \widetilde{b}_j$, $\widetilde{a} = a_0 + \widetilde{b}$, and $\widetilde{B} = h + \widetilde{b}^T f$. By the product rule and (S10),

$$\widetilde{e}_B = -\{\widetilde{b}_\tau + D\widetilde{b} f + Df^T \widetilde{b} + (r, 0)\}, \qquad \widetilde{f}_a = f - \widetilde{K}^{-1} \widetilde{e}_B, \quad \widetilde{K} = K_0 + D\widetilde{b}^T - D\widetilde{b}. \tag{S17}$$

At the center, exact coefficient values and first derivatives reproduce the native time-jet residual. Away from the center, approximation of these coefficients causes a separate error. At all points $d\widetilde{\Omega} = 0$ holds algebraically, whereas time compatibility with the original field must be checked independently. Exact compatibility holds for $\widetilde{f}_a$ by its defining representation.

Differentiating $\widetilde{K}\widetilde{f}_a = \nabla\widetilde{B} + \partial_\tau \widetilde{a}$ in the coordinate z_j and collecting columns yields

$$D\widetilde{f}_a = \widetilde{K}^{-1} \left(\nabla^2 \widetilde{B} + D\widetilde{b}_\tau - [\partial_{z_j} \widetilde{K}\, \widetilde{f}_a]_j \right). \tag{S18}$$

Every derivative of the actual force and Hamiltonian occurring in $\widetilde{B}$ belongs in this expression. In particular, differentiating a closed-loop field includes the state derivative of its feedback. Freezing feedback in this tangent would differentiate a different field.

1.4 Moser pullback, state sensitivities, and domain control

Fix τ and put $\Omega_{\lambda,\tau} = \Omega_0 + \lambda d(\widetilde{b}^T dz)$. Assume its matrix $K_{\lambda,\tau}$ is nonsingular for $0 \le \lambda \le 1$, and the flow of $v_{\lambda,\tau} = K_{\lambda,\tau}^{-1}\widetilde{b}$ exists throughout this interval in a common domain. Then $\iota_v \Omega_{\lambda,\tau} = -\widetilde{b}^T dz$, and

$$\frac{d}{d\lambda}\rho_{\lambda,\tau}^* \Omega_{\lambda,\tau} = \rho_{\lambda,\tau}^* \{ d(\widetilde{b}^T dz) + \mathcal{L}_v \Omega_{\lambda,\tau} \} \tag{S19}$$

$$= \rho_{\lambda,\tau}^* \{ d(\widetilde{b}^T dz) + d\iota_v \Omega_{\lambda,\tau} \} = 0. \tag{S20}$$

Thus $\rho_{1,\tau}^* \widetilde{\Omega}_\tau = \Omega_0$, and for $\chi_\tau = \rho_{1,\tau}^{-1}$, $D\chi_\tau^T K_0 D\chi_\tau = \widetilde{K}$. This establishes the exact auxiliary-flow identity. Finite numerical integration has a separately measured pullback defect.

If $z = \rho(y,\tau)$, where $\rho = \rho_{1,\tau}$, differentiation along a represented trajectory gives $\widetilde{f}_a = \rho_\tau + D_y \rho\, \dot{y}$. Therefore $u = \dot{y} = \mathsf{R}^{-1}(\widetilde{f}_a \circ \rho - \upsilon)$, with $\mathsf{R} = D_y\rho$ and $\upsilon = \rho_\tau$. Compatibility of the represented form and the coordinate transformation implies $\mathcal{L}_u \Omega_0 = 0$. Consequently $d((K_0 u)^T dy) = 0$. On a star-shaped domain about y_*,

$$\langle(y,\tau) = \int_0^1 (K_0 u)(y_* + a(y - y_*), \tau)^T (y - y_*)\, da \tag{S21}$$

has gradient $K_0 u$ by the radial Poincaré formula, and $u = J\nabla\langle$. No formula for this scalar justifies dropping υ.

Differentiating $\rho' = v(\rho, \lambda, \tau)$ first in initial state and physical time gives

$$\mathsf{R}' = Dv\mathsf{R}, \qquad \upsilon' = Dv\upsilon + v_\tau, \qquad \mathsf{R}(0) = I, \quad \upsilon(0) = 0. \tag{S22}$$

Differentiating again gives, with zero initial second sensitivities,

$$\mathsf{S}' = Dv\mathsf{S} + D^2 v[\mathsf{R},\mathsf{R}], \qquad \mathsf{W}' = Dv\mathsf{W} + D^2 v[\mathsf{R},\upsilon] + Dv_\tau \mathsf{R}, \tag{S23}$$

where $\mathsf{S} = D_y^2 \rho$, $\mathsf{W} = D_y \rho_\tau$. Finally differentiate $\mathsf{R}u = \widetilde{f}_a \circ \rho - \upsilon$ in an arbitrary direction ζ:

$$Du[\zeta] = \mathsf{R}^{-1}\{ D\widetilde{f}_a \mathsf{R}\zeta - \mathsf{W}\zeta - \mathsf{S}[\zeta, u] \}. \tag{S24}$$

This is the full moving-chart tangent, not a frozen-chart derivative.

For the quadratic coefficient cache, $\widetilde{K}$ is affine in z. On a radius-R_b ball about z_*, the triangle and Cauchy–Schwarz inequalities give

$$\delta_K = \|\widetilde{K}(z_*) - K_0\|_2 + R_b \left(\sum_j \|\partial_{z_j} \widetilde{K}\|_2^2 \right)^{1/2}, \tag{S25}$$

$$b_{\max} = \|\widetilde{b}(z_*)\| + R_b \|D\widetilde{b}(z_*)\|_2 + \tfrac{1}{2} R_b^2 \|D^2 \widetilde{b}\|_F. \tag{S26}$$

Since all singular values of K_0 equal one, $\delta_K < 1$ implies $\sigma_{\min}(K_{\lambda,\tau}) \ge 1 - \delta_K$, uniformly in the ball and λ. Hence $\|v\| \le b_{\max}/(1 - \delta_K)$. If the initial distance from z_* plus this speed bound is strictly less than R_b, a first-exit argument prevents the exact auxiliary trajectory from reaching the boundary before $\lambda = 1$. These are analytical sufficient conditions for the specified polynomial. Floating-point evaluation without directed rounding does not turn them into an interval-certified bound for all physical coefficients.

1.5 Proof of Theorem A and the initial gauge

Fix endpoint times s, τ, with closed, nondegenerate forms Ω_s and Ω_τ on the respective endpoint domains. Let Φ be a local diffeomorphism preserving the specified two time sections. Suppose a locally invertible joint endpoint transformation $\mathcal{E}$ satisfies $\mathcal{E}^* d(\widehat{w}^T dw) = \Omega_\tau(z_+) - \Omega_s(z_-)$. For the graph embedding $\iota : z_- \mapsto (\Phi(z_-), z_-)$,

$$d\,\iota^* \mathcal{E}^* (\widehat{w}^T dw) = \Phi^* \Omega_\tau - \Omega_s = 0. \tag{S27}$$

Write $D\mathcal{E} = \left(\begin{smallmatrix} A & B_e \\ C & D_e \end{smallmatrix}\right)$ and $M_\Phi = D\Phi$. On this graph,

$$d\widehat{w} = (AM_\Phi + B_e)\, dz_-, \qquad dw = (CM_\Phi + D_e)\, dz_-. \tag{S28}$$

If $CM_\Phi + D_e$ is nonsingular, the inverse function theorem makes w a coordinate on a restricted graph neighborhood. Restrict further to a star-shaped coordinate domain. Closedness in (S27) and the Poincaré lemma then provide $\widehat{w} = \nabla\phi$, and

$$N = D_w\widehat{w} = (AM_\Phi + B_e)(CM_\Phi + D_e)^{-1} = N^T. \tag{S29}$$

Matrix order follows directly from (S28) and cannot be interchanged.

Conversely, let ϕ be a local scalar and define $z_\pm(w)$ through the full joint inverse $\mathcal{E}^{-1}(\nabla\phi, w)$. If $D_w z_-$ is nonsingular, these endpoints form the graph of a local map Ψ. Pullback of $d(\nabla\phi^T dw) = 0$ through this graph and endpoint compatibility give $\Psi^*\Omega_\tau = \Omega_s$. This proves the reverse implication and identifies its extra projection condition. The relation $N(CM_\Phi + D_e) = AM_\Phi + B_e$ can, when its coefficient is invertible, be rearranged as

$$M_\Phi = (A - NC)^{-1}(ND_e - B_e). \tag{S30}$$

It does not involve N^{-1}. For central endpoints at the identity map, $N = 0$ and $CM_\Phi + D_e = I$, so demanding an invertible generator Hessian would exclude a valid initial state.

At $\tau = s$, define $w_0(z) = \mathcal{E}_2(z, z, s, s)$. When $D_z w_0 = C_0 + D_{e,0}$ is invertible, set $z = z(w)$ and $c_0(w; s) = \mathcal{E}_1(z(w), z(w), s, s)$. The diagonal graph makes the endpoint two-form difference zero, so Dc_0 is symmetric. Let $x = w - w_*$. The normalized radial integral

$$\phi_0(w; s) = \int_0^1 c_0(w_* + ax; s)^T x\, da \tag{S31}$$

has derivative in direction ζ equal to

$$\int_0^1 \{a(Dc_0\zeta)^T x + c_0^T\zeta\}\, da = \int_0^1 \frac{d}{da}\{ac_0(w_* + ax; s)^T\zeta\}\, da = c_0(w; s)^T\zeta. \tag{S32}$$

It also vanishes at w_*. Symmetry of Dc_0 is the needed step; radial integration of an arbitrary nonclosed vector field does not produce its potential. General endpoint gauges can have $c_0 \neq 0$: replacing $\widehat{w} = J(z_+ - z_-)$ by $J(z_+ - z_-) + \nabla\psi(w)$ preserves the endpoint two-form but yields $c_0 = \nabla\psi$. The zero initial generator is therefore special to the ungauged central construction.

An initial positive smallest singular value ensures transversality only nearby. If $T = CM_\Phi + D_e$ and $\|T(\tau, w) - T(s, w_*)\|_2 < \sigma_{\min}(T(s, w_*))$ on a chosen neighborhood, the singular-value perturbation inequality guarantees regularity there. An analogous condition is needed for the negative-endpoint projection in reconstruction. Continuation ends if the joint inverse fails, a projection becomes singular, or an endpoint leaves the common domain. Sampled condition numbers do not establish these uniform neighborhood bounds.

1.6 Proof of Theorem B and scalar central recursion

Keep s and the initial endpoint fixed, and differentiate the moving final endpoint. At the point reconstructed by $\mathcal{E}^{-1}(g, w, \tau, s)$,

$$\dot{\widehat{w}} = Af(z_+, \tau) + \mathcal{E}_{1,\tau}, \qquad \dot{w} = Cf(z_+, \tau) + \mathcal{E}_{2,\tau}. \tag{S33}$$

Since $\widehat{w} = g(w, \tau; s)$, the chain rule yields $\dot{\widehat{w}} = D_w g\, \dot{w} + \partial_\tau g|_w$. Subtracting the second contribution gives

$$\partial_\tau g|_w = (A - NC)f + \mathcal{E}_{1,\tau} - N\mathcal{E}_{2,\tau} = \Psi(g, w, N, \tau; s), \qquad N = D_w g. \tag{S34}$$

The explicit time derivatives include any time-dependent charts used to build $\mathcal{E}$. Differentiation along the moving midpoint is not a replacement for this fixed-w derivative.

Assume analyticity of the field, forms, and endpoint transformation on common local neighborhoods, with analytic joint inverse and regular projections. Analytic flow dependence and the analytic implicit function theorem then give an actual local analytic g. Set $g = \sum_{j\geq 0} \theta^j c_j$. The coefficient of θ^m in its time derivative is $(m+1)c_{m+1}$. The coefficient on the right depends only on $c_0, \ldots, c_m$ and their w derivatives, so

$$c_{m+1} = \frac{1}{m+1}[\theta^m]\Psi\left(\sum_{j=0}^{m}\theta^j c_j, w, \sum_{j=0}^{m}\theta^j D_w c_j, s+\theta; s\right). \tag{S35}$$

In particular $c_1 = \Psi_0$ and $2c_2 = (\Psi_\tau + \Psi_g c_1 + \Psi_N[D_w c_1])_0$. The last term is a Fréchet derivative with respect to the matrix argument. It cannot be omitted. Symmetry of the analytic $D_w g$ implies coefficientwise symmetry of every $D_w c_j$, so radial integration on the common star-shaped domain supplies $c_j = \nabla\phi_j$. A finite sum remains an exact gradient, and Theorem A reconstructs a structure-preserving local map if its negative projection remains invertible.

If the truncated gradient error is bounded by $C\eta^{r_o+1}$ and the endpoint inverse and negative projection inverses have uniform derivative bounds, the mean value theorem and the implicit function theorem transfer this bound to the endpoint state error. A one-step stability factor $1 + C\eta$ then gives global order r_o on fixed time intervals by summation. Additional symmetry can cancel the leading defect. This reasoning uses a local state defect, not merely the number of nonzero displayed coefficients. With finite smoothness one replaces analyticity by a finite Taylor theorem of the required order and makes no infinite-series claim.

For canonical endpoints choose $\widehat{w} = J(y_+ - y_-)$ and $w = (y_+ + y_-)/2$. Then

$$y_+ = w - \tfrac{1}{2}Jg, \quad y_- = w + \tfrac{1}{2}Jg, \qquad A = J, \quad C = I/2. \tag{S36}$$

For $u = J\nabla\langle$, equation (S34) becomes $(J - N/2)J\nabla\langle(y_+, \tau)$. On the other hand, because $N = N^T$,

$$\nabla_w[-\langle(w - Jg/2, \tau)] = -(I + NJ/2)\nabla\langle = (J - N/2)J\nabla\langle. \tag{S37}$$

The two scalar sides therefore differ by a function only of time. Choosing that time gauge gives

$$\partial_\tau\phi = -\langle(w - J\nabla\phi/2, \tau), \qquad \phi(w, s; s) = 0, \tag{S38}$$

and coefficient extraction gives the computable scalar recursion

$$\phi_{m+1} = -\frac{1}{m+1}[\theta^m]\langle\left(w - \tfrac{1}{2}J\nabla\sum_{j=1}^{m}\theta^j\phi_j, s+\theta\right). \tag{S39}$$

For an autonomous Hamiltonian, $\phi_1 = -\langle$, $\phi_2 = 0$. To find ϕ_3, the shifted argument is $w + \theta J\nabla\langle/2 + O(\theta^3)$; the linear contribution vanishes since $\nabla\langle^T J\nabla\langle = 0$. The quadratic coefficient is $f^T\langle'' f/8$, giving $\phi_3 = -f^T\langle'' f/24$. Reversibility makes the local central generator odd in the step, so $\phi_4 = 0$. General explicit time dependence removes this oddness; for $\langle = h + \tau Q^2/2$ at $s = 0$, the next coefficient is $\phi_2 = -Q^2/4$. This scalar example illustrates a restriction on extrapolating the autonomous formula, rather than an implemented general nonautonomous 24-dimensional recursion.

1.7 The original endpoint equations and noncommuting matrix elimination

For endpoint order $Z = (Q_+, P_+, Q_-, P_-)$, introduce $(\widehat{\xi}, \widehat{\pi}, \xi, \pi)$ with $\xi = \overline{Q}$, $\pi = -\overline{P}$, $\widehat{\xi} = \delta P$, and $\widehat{\pi} = \delta Q$. The joint linear transformation and inverse are

$$L = \begin{pmatrix} 0 & I & 0 & -I \\ I & 0 & -I & 0 \\ I/2 & 0 & I/2 & 0 \\ 0 & -I/2 & 0 & -I/2 \end{pmatrix}, \qquad L^{-1} = \begin{pmatrix} 0 & I/2 & I & 0 \\ I/2 & 0 & 0 & -I \\ 0 & -I/2 & I & 0 \\ -I/2 & 0 & 0 & -I \end{pmatrix}. \tag{S40}$$

Block multiplication gives both products equal to the identity. Fixing the negative endpoint and differentiating gives

$$\dot{\widehat{w}} = \begin{pmatrix} \dot{P}_+ \\ \dot{Q}_+ \end{pmatrix}, \qquad \dot{w} = \tfrac{1}{2}\begin{pmatrix} \dot{Q}_+ \\ -\dot{P}_+ \end{pmatrix}, \qquad \partial_\tau g|_w = \dot{\widehat{w}} - N\dot{w}. \tag{S41}$$

The signs and block positions follow from L. The scalar $-\eta h(\xi, -\pi, \tau_m)$ has gradient $\eta(-h_Q, h_P)$, producing the conservative midpoint endpoint difference. For a damped oscillator $\dot{P} = -Q - \gamma P$, the apparent gradient in these variables would be $(-\xi + \gamma\pi, -\pi)$. Its cross derivatives are γ and zero. Hence, for $\gamma \neq 0$, it cannot be the gradient of a local scalar in this unmodified canonical endpoint chart. Compatible time-dependent forms or a valid specialization are necessary.

For positive-definite $M(Q)$, differentiating $M^{-1}M = I$ gives $\partial_i M^{-1} = -M^{-1}(\partial_i M)M^{-1}$. Thus Hamilton's equations with generalized load give the full midpoint equations

$$\delta Q = \eta M(\overline{Q})^{-1}\overline{P}, \qquad \delta P_i = \eta\{-\tfrac{1}{2}\overline{P}^T \partial_i M^{-1}(\overline{Q})\overline{P} - V_{,i}(\overline{Q}) + r_i(\overline{z}, \tau_m)\}. \tag{S42}$$

All terms use the same unknown midpoint. This preserves the configuration derivatives of the kinetic energy. When M, K_{lin}, and C_d are constant and the force is prescribed, the equations reduce to

$$\delta Q = \eta M^{-1}\overline{P}, \qquad \delta P = -\eta K_{\text{lin}}\overline{Q} - \eta C_d M^{-1}\overline{P} + \eta F_m. \tag{S43}$$

The first equation gives $\overline{Q} = Q_- + \eta M^{-1}(P_+ + P_-)/4$. Substitution into the second, with $\overline{P} = (P_+ + P_-)/2$, yields

$$(I + E)P_+ = (I - E)P_- - \eta K_{\text{lin}} Q_- + \eta F_m, \qquad E = \tfrac{\eta^2}{4} K_{\text{lin}} M^{-1} + \tfrac{\eta}{2} C_d M^{-1}. \tag{S44}$$

The multiplication orders follow from substitution and require no commutation assumption. If $I + E$ is nonsingular, solve this left system and recover $Q_+ = Q_- + \eta M^{-1}(P_+ + P_-)/2$. A midpoint feedback force depending on the unknown endpoints still leaves an implicit equation. For variable mass, this constant-matrix elimination is unavailable.

For any smooth field, the complete time-midpoint scheme has local defect $O(\eta^3)$ by Taylor expansion about the temporal midpoint, including explicit time derivatives. Its global second order follows under the stability conditions proved below. Calling the generating approximation "first degree in the step" does not reduce the order of this symmetric time-midpoint update. A general damped midpoint remains a second-order approximation without automatically preserving the canonical form.

1.8 Nonseparable Central4, symmetry, and the exact primitive

Let $f = J\nabla h$ be autonomous, with sufficiently many bounded derivatives on the common domain for the displayed remainders. Let z_c be the exact solution at the temporal midpoint. Taylor expansion at $\pm\eta/2$ gives

$$m = z_c + \frac{\eta^2}{8} f' f(z_c) + O(\eta^4), \tag{S45}$$

$$z_+ - z_- = \eta f(z_c) + \frac{\eta^3}{24}\{f''[f, f] + (f')^2 f\}(z_c) + O(\eta^5). \tag{S46}$$

The inverse relation is $z_c = m - \eta^2 f' f(m)/8 + O(\eta^4)$. Substituting it into the leading field term changes its cubic contribution by $-\eta^3 (f')^2 f/8$, whereas the already cubic term can be evaluated at m up to $O(\eta^5)$. Hence

$$z_+ - z_- = \eta f(m) + \frac{\eta^3}{24}\{f''[f, f] - 2(f')^2 f\}(m) + O(\eta^5). \tag{S47}$$

For $A_h = h''$ and $q_h = f^T A_h f$, differentiation in a direction ζ gives

$$Dq_h[\zeta] = 2(Df\zeta)^T A_h f + h^{(3)}[\zeta, f, f] \tag{S48}$$

$$= \zeta^T\{h^{(3)}[f, f] - 2A_h J A_h f\}, \tag{S49}$$

because $Df = JA_h$ and $(JA_h)^T = -A_h J$. Multiplying the gradient by J yields $J\nabla q_h = f''[f,f] - 2(f')^2 f$. Therefore

$$S_\eta = \eta h + \frac{\eta^3}{24} q_h, \qquad \delta z = J\nabla S_\eta(m) \tag{S50}$$

matches the exact endpoint relation through order four, with local state defect $O(\eta^5)$ on a uniformly regular implicit branch. The generator satisfies $S_{-\eta} = -S_\eta$, so interchanging the endpoints and reversing the step leaves the equation unchanged. It is thus symmetric. No part of this derivation assumes a separable Hamiltonian; all mixed mass and potential derivatives remain in h'' and $h^{(3)}$.

To prove symplecticity independently, differentiate the implicit equation. With $C_\eta = JS''_\eta(m)/2$,

$$D\Phi = (I - C_\eta)^{-1}(I + C_\eta), \qquad C_\eta^T K_0 + K_0 C_\eta = 0. \tag{S51}$$

The second relation follows from the symmetry of S''_η. Expanding both sides gives $(I+C_\eta)^T K_0 (I+C_\eta) = (I - C_\eta)^T K_0 (I - C_\eta)$. Since $I - C_\eta$ and $I + C_\eta$ commute, multiplication by the inverse factors proves $D\Phi^T K_0 D\Phi = K_0$.

Exact symplecticity requires the stronger primitive statement. Write $\lambda_0 = P^T dQ$ and expand $P_\pm = \overline{P} \pm \delta P/2$, $Q_\pm = \overline{Q} \pm \delta Q/2$. The terms even in the endpoint differences cancel, giving

$$\begin{aligned} P_+^T dQ_+ - P_-^T dQ_- &= \overline{P}^T d(\delta Q) + \delta P^T d\overline{Q} && \text{(S52)} \\ &= \overline{P}^T d(\delta Q) - S_Q^T d\overline{Q} && \text{(S53)} \\ &= d\{\overline{P}^T \delta Q - S_\eta(\overline{Q}, \overline{P})\}, && \text{(S54)} \end{aligned}$$

where the last equality uses $\delta Q = S_P$ to cancel the $d\overline{P}$ terms. This proves exactness wherever the generator and implicit branch are single-valued. Ordinary midpoint is the case $S_\eta = \eta h$. The primitive argument concerns the ideal implicit map. It is not an identity for arbitrary finite-iteration output.

The distinction between closed and exact matters on a torus. The closed form $\sum_j I_{0,j} d\theta_j$ has basic periods $2\pi I_{0,j}$ and need not be a differential of a single-valued scalar. More generally, if a symplectic tube map F on $\mathbb{T}^n \times U_I$, with contractible U_I, gives $\beta = F^*\lambda_0 - I^T d\theta$, define $I_{0,j} = (2\pi)^{-1} \oint_{\gamma_j} \beta$. Subtracting these periods leaves a closed form with zero periods and hence a single-valued primitive:

$$F^*\lambda_0 = (I + I_0)^T d\theta + dS. \tag{S55}$$

This identity specifies a shifted reference primitive; it does not silently set I_0 to zero. The local Central4 primitive above has its own direct proof and does not depend on such a tube-map assumption.

1.9 Uniform damping and the scope of a weighted midpoint

Suppose the sole additional load is $r = -\gamma P$ with constant γ. Define $\Pi = e^{\gamma\tau} P$ and $\langle((Q, \Pi, \tau) = e^{\gamma\tau} h(Q, e^{-\gamma\tau}\Pi)$. Then

$$\dot{Q} = h_P = \langle_\Pi, \qquad \dot{\Pi} = \gamma e^{\gamma\tau} P + e^{\gamma\tau}(-h_Q - \gamma P) = -\langle_Q. \tag{S56}$$

This proof allows configuration-dependent mass. Center the time coordinate at the step midpoint and write the lifted endpoint momenta as $\Pi_- = e^{-\gamma\eta/2} P_-$, $\Pi_+ = e^{\gamma\eta/2} P_+$. At this midpoint the lifted Hamiltonian equals h. Canonical time-midpoint therefore gives

$$\Phi_\eta = S_{\eta/2} \circ \mathrm{MP}_{h,\eta} \circ S_{\eta/2}, \qquad S_a(Q,P) = (Q, e^{-\gamma a} P). \tag{S57}$$

The scaling and its composition with symplectic midpoint satisfy

$$\begin{aligned} DS_a^T K_0 DS_a &= e^{-\gamma a} K_0, \\ D\Phi_\eta^T K_0 D\Phi_\eta &= e^{-\gamma\eta} K_0. \end{aligned} \tag{S58}$$

Equivalently, the update preserves the two time sections $e^{\gamma\tau} K_0$. The lifted complete time-midpoint method is second order. Anisotropic loads, state-dependent feedback, or contact switching do not satisfy the hypothesis used in this calculation.

1.10 Finite-time stability, algebraic residuals, and representation error

Consider a smooth field on an open domain containing a closed convex neighborhood, with spatial Lipschitz constant L, and the midpoint residual

$$\mathcal{F}_k(x) = x - z_k - \eta f((x + z_k)/2, \tau_k + \eta/2). \tag{S59}$$

Assume its fixed-point map stays in that neighborhood and $a_\eta = \eta L/2 < 1$. Banach's theorem gives a unique fixed point there. If x_* is the exact root and $\mathcal{F}_k(x) = \varepsilon_k$, subtraction and the Lipschitz bound give

$$(1 - a_\eta)\|x - x_*\| \le \|\varepsilon_k\|. \tag{S60}$$

For two different inputs, subtracting their exact midpoint equations yields the step stability factor $q_\eta = (1 + a_\eta)/(1 - a_\eta)$. Comparing a numerical step with the exact solution, whose local increment defect is $O(\eta^3)$, gives

$$e_N \le q_\eta^N e_0 + \frac{1}{1 - a_\eta} \sum_{j=0}^{N-1} q_\eta^{N-1-j}(C\eta^3 + \|\varepsilon_j\|). \tag{S61}$$

For $N\eta \le T$ and a_η uniformly below one, $q_\eta^N \le e^{C_L T}$. Summing N local defects therefore gives $e_N \le C_T(e_0 + \eta^2 + \sum_j \|\varepsilon_j\|)$. If all residuals are at most ε_η, their contribution is $O(\varepsilon_\eta/\eta)$. A code tolerance in the infinity norm can be converted to the Euclidean norm by at most $\sqrt{2n}$. A fixed absolute solver tolerance cannot retain convergence order under arbitrarily fine step refinement.

For Central4, $S''_\eta = O(\eta)$ on a bounded derivative domain. A uniform inverse bound for $I - JS''_\eta/2$ and step stability $1 + O(\eta)$ produce the same argument with local defect $O(\eta^5)$. Thus the global discretization term is $O(\eta^4)$, with the same residual sum. For the weighted method and exact Moser charts, bounded chart derivatives and inverse derivatives on the finite interval transfer the state and residual bounds between coordinates. If finite numerical chart operations introduce additional per-step state errors δ_j, stability accumulates $\sum_j \delta_j$ with the same finite-time weights. Exact structure of the ideal map alone does not bound these implementation errors.

Finally compare solutions of $\dot z = f(z, \tau)$ and $\dot z_a = \widetilde{f}_a(z_a, \tau)$ starting at the same point. If both remain in a common domain, f is Lipschitz with constant L, and $\|\widetilde{f}_a - f\| \le \epsilon_f$ uniformly there, then

$$\|z_a(\tau) - z(\tau)\| \le \int_0^\tau \{L\|z_a(s) - z(s)\| + \epsilon_f\}\, ds \le \epsilon_f \frac{e^{L\tau} - 1}{L}, \tag{S62}$$

with value $\epsilon_f \tau$ at $L = 0$. Adding this bound to the integration estimate yields the conditional total finite-time bound. The uniform field difference contains time truncation and spatial coefficient approximation, while numerical chart and implicit-solve errors enter their own sums. A fixed-time coefficient residual of order θ^{r_o} is not itself a statement of order r_o for the chart midpoint integrator. Reducing the numerical step can converge to the represented trajectory while leaving a nonzero discrepancy from the original trajectory. No pointwise numerical diagnostic establishes the uniform assumptions of this theorem or the separate hypotheses needed for a continuous or discrete KAM result.

1.11 Derivative evaluation and the limit of computational verification

The identity $h_{\mathrm{AD}}(Q, P) = P^T M(Q)^{-1} P/2 + h(Q, 0)$ preserves the full mechanical Hamiltonian, including its variable mass. It supplies an alternative computational expression, not a theorem that any automatic differentiation path returns correct high derivatives. The scalar example $h(Q, P) = P^2/[2(1 + Q^2)]$ has $h_{QQPP}(0) = h_{PPQQ}(0) = -2$ by expansion. The recorded mixed differentiation path through a linear solve returned zero for one fourth-order component in the tested software version. Full tensor symmetrization cannot generally repair this, because contraction against four copies of one vector is unchanged by permutation averaging.

The retained implementation checks selected third- and fourth-order contractions against independent pure reverse differentiation and finite differences, and uses the inverse expression for the tested transport

paths. The physical reference field retains the original solve-based evaluator. Central4 differentiates its complete scalar generator through a separate reverse-mode path. These checks support the specified version, points, and directions; they do not prove all-state, all-order derivative correctness. The analytical arguments above concern the mathematical coefficients, with computational defects remaining part of the numerical evidence and error budget.

2 Learning structure, error bounds and persistence scope

2.1 Implemented architectures and inverse diagnostics

For clarity, the complex network's input is formed from training coordinates: $\zeta = (x_Q + ix_P)/\sqrt{2}$, $u = (\zeta, \bar{\zeta}) \in \mathbb{C}^{24}$. With componentwise products denoted by $\odot$, its implemented residual is

$$\begin{aligned} a_1 &= W_1 u, \\ b &= \sigma(W_2 \sigma(a_1) + b_2), \\ F(x) &= \mathrm{Re}\{w_3 (a_1^{\odot 3} \odot b)\}, \\ r_\theta(x) &= \tfrac{s_h}{2}\{F(x_Q, x_P) + F(x_Q, -x_P)\}. \end{aligned} \tag{S63}$$

The two hidden widths are 48; first and output layers have no biases. Parameter counts are $1152 + 2304 + 48 + 48 = 3552$ complex entries. Because the first layer vanishes at the origin, the cubic gate has vanishing derivatives through order two there. Momentum averaging gives evenness, but permits higher even momentum powers.

The SPD model replaces u by the real position input x_Q to a width-48 complex trunk. Its mass head is $r = \mathrm{Re}\{W_m[b(x_Q) - b(0)]\} \in \mathbb{R}^{78}$. Fill the lower triangle of L with these entries, exponentiating the diagonal; use $V_\theta = s_h \,\mathrm{Re}\{w_V (a_1^{\odot 3} \odot b)\}$ for the potential. This gives $576 + 2304 + 48 + 3744 + 48 = 6720$ complex entries, and $L(0) = I$. Thus the exact momentum Hessian is positive definite at each finite real input, while the potential and full Hessian need not be convex.

The Legendre test uses SciPy HYBR with tolerance 10^{-10} and the setting `maxfev=120`, initialized at the true momentum. Table S1 records maxima across the three seeds and eight prescribed states per split. Polynomial and SPD equation residuals are below 10^{-8} at all sampled points; their overall maxima are 2.15×10^{-14} and 1.80×10^{-15}. Some progress flags are false despite these residuals. For the unrestricted complex model, residual convergence counts are 4/6/5 out of eight test states and 3/4/6 out of eight wider-domain states; maximum residuals are 0.0434103 and 0.521317. Equation satisfaction, solver status and recovery of physical momentum are therefore distinct measurements.

Table S1: Maximum sampled relative momentum-Hessian error, Hessian condition number and relative recovered-momentum error. Each entry is test/wider domain.

Model	Relative h_{PP} error	Condition number	Relative momentum error
Polynomial	0.4749/0.7316	22124/3480	27.30/7.244
Complex	0.05819/0.2917	167322/41387	0.6267/1.348
SPD complex	0.1705/0.3132	1199/1811	0.5044/1.253

The historical SPD stress-domain experiment reported Hessian errors of 320%/372% and condition numbers approximately $4.61 \times 10^{10}/1.13 \times 10^{12}$. These belong to earlier weights whose original numerical archive is absent from the snapshot; they are historical evidence, not remeasured values or current-model statistics. They illustrate the same distinction between exact structural positivity and usable conditioning. The earlier meromorphic-activation failure is likewise retained in the source history.

The learning-to-coil diagnostic uses four test indices $(0, 146, 292, 511)$ and four wider-domain indices $(0, 72, 144, 255)$. Allocation constraints are 2 A, 20 A/s, 5 mT and gradient Frobenius norm 0.03 T/m at the actual magnet quadrature points, with a preenergized bias and a 0.25 s decision interval. The maximum final constraint violation is 9.39×10^{-13}. Unconstrained requested peak currents reach 326.421 A,

37.9885 A and 235.468 A for polynomial, complex and SPD models. 23 failures of the initial allocation attempt were preserved; the final 72/72 success result uses a feasible previous-current warm start for the same force objective without relaxing limits. This experiment assesses correction through the coil port; it does not equate the ideal-bias and coil Hamiltonians.

2.2 Deterministic finite-feature error bound

Let $h_\theta = h_0 + \sum_j c_{\theta j}\psi_j$, $h_* = h_0 + \sum_j c_{*j}\psi_j + R$, and $Y_i = \nabla h_*(z_i) + \eta_i$. Remove constant basis directions and stack the derivative-feature matrices into A. For one fixed scalar-observation normalization N_o, assume

$$\begin{aligned} G &= A^T A/N_o \succeq \lambda I, \qquad \lambda > 0, \\ r &= \|\nabla h_\theta - Y\|_2/\sqrt{N_o}, \\ e &= \|\nabla R + \eta\|_2/\sqrt{N_o}. \end{aligned} \tag{S64}$$

At the samples, $A\Delta c = (\nabla h_\theta - Y) + (\nabla R + \eta)$ for $\Delta c = c_\theta - c_*$. Hence

$$\sqrt{\lambda}\|\Delta c\|_2 \le (\Delta c^T G \Delta c)^{1/2} \le r + e. \tag{S65}$$

For any function norm obeying the triangle inequality, Cauchy–Schwarz then gives

$$\begin{aligned} \|h_\theta - h_*\|_{\mathcal{X}} \le \Big(\sum_j \|\psi_j\|_{\mathcal{X}}^2\Big)^{1/2} \frac{r+e}{\sqrt{\lambda}} \\ + \|R\|_{\mathcal{X}}. \end{aligned} \tag{S66}$$

The proof concerns the fitted coefficient vector itself and does not require an optimizer-specific convergence claim. For monomials $\psi = z^\alpha$, the analytic norm on $|z_j| \le \rho$ is at most $\rho^{|\alpha|}$, and a real C^ℓ bound is

$$\max_{|\beta|\le\ell,\, \beta\le\alpha} \frac{\alpha!}{(\alpha-\beta)!}\rho^{|\alpha|-|\beta|}. \tag{S67}$$

An analytic action–angle chart must map into the certified complex domain. A real C^ℓ composition instead needs bounds for the chart derivatives. For integrated equations, the same algebra applies to the actual integrated feature matrix; quadrature, endpoint and observation errors enter e. Export rounding adds a separate norm bound. A small observed residual does not bound the out-of-class term R.

2.3 Conditional statistical strong-norm bound

Let $\mathcal{D} = \mathbb{T}^n \times B$ have smooth bounded action domain, $d = 2n$, $p > d$, and integers $\ell \ge 1$, $s > \ell + d/p$. Let X_i be independent with density $f \ge m_0 > 0$. Realify every trainable complex parameter and assume $\Theta \subset [-B_\Theta, B_\Theta]^{P_\Theta}$. After subtracting the spatial mean, set $e_\theta = h_\theta - h_* - \overline{h_\theta - h_*}$. Assume throughout the domain and parameter class that

$$\begin{aligned} \|\nabla e_\theta(x)\|_p &\le Q_*, \\ \|\nabla h_\theta(x) - \nabla h_\eta(x)\|_p &\le L_g\|\theta - \eta\|_\infty, \\ \|e_\theta\|_{W^{s,p}} &\le M_s. \end{aligned} \tag{S68}$$

These are quantitative class assumptions. A parameter Lipschitz constant in Euclidean norm must first be multiplied by $\sqrt{P_\Theta}$ to use the displayed infinity norm.

For fixed covering radius $0 < b \le 2B_\Theta$, an infinity-norm net has at most $(1 + 2B_\Theta/b)^{P_\Theta}$ points. The loss $r_\theta(x) = \|\nabla e_\theta(x)\|_p^p$ lies in $[0, Q_*^p]$ and is $L_{\text{par}} = pQ_*^{p-1}L_g$-Lipschitz in parameters. Hoeffding and the union bound control all net points with probability at least $1 - \delta$. Extending both empirical and population

losses from the net costs $2L_{\text{par}}b$, giving simultaneously

$$\begin{aligned} R_\theta &\le \widehat{R}_\theta + \Gamma, \\ \Gamma &= 2L_{\text{par}}b \\ &\quad + Q_*^p \sqrt{\frac{P_\Theta \log(1 + 2B_\Theta/b) + \log(1/\delta)}{2N}}, \end{aligned} \tag{S69}$$

where $\widehat{R}_\theta = N^{-1}\sum_i r_\theta(X_i)$ and $R_\theta = \int r_\theta \, d\mu$. The event therefore includes parameters selected using these training data.

The density bound and Poincaré inequality for mean-zero functions imply

$$\begin{aligned} \|e_\theta\|_{W^{1,p}} &\le G_\theta, \\ G_\theta &:= C_P m_0^{-1/p} (\widehat{R}_\theta + \Gamma)^{1/p}. \end{aligned} \tag{S70}$$

Apply Gagliardo–Nirenberg to $g = \nabla e_\theta$, with derivative orders $j = \ell - 1$, $m = s - 1$ and both endpoint integrability exponents equal to p. Solving

$$\begin{aligned} 0 &= \frac{j}{d} + \alpha\left(\frac{1}{p} - \frac{m}{d}\right) + \frac{1-\alpha}{p} \\ \Longrightarrow \quad \alpha &= \frac{\ell - 1 + d/p}{s-1} \in (0,1). \end{aligned} \tag{S71}$$

and using extension from the action domain and periodic angular charts yields

$$\|e_\theta\|_{C^\ell} \le C_{\text{GN}} \left(G_\theta + M_s^\alpha G_\theta^{1-\alpha}\right). \tag{S72}$$

The inhomogeneous term and $W^{1,p} \hookrightarrow C^0$ control lower derivatives. This completes the conditional implication. Neither trajectory-adjacent samples nor a low empirical loss supply the required sampling, coverage and uniform high-derivative constants.

2.4 Conditional persistence of the exact discrete map

Let $h \circ \mathcal{T} = h_0 + P$ with $\|P\| \le \varepsilon$ on a declared complex action–angle domain. Require analytic canonical $\mathcal{T}$ and inverse, controlled derivatives, an analytic exact-symplectic step Ψ_Δ and inverse on retained domains, and a unique implicit branch whose endpoints remain there. For a central generator S_Δ, $\frac{1}{2}\sup\|JS_\Delta''\| < 1$ is a sufficient local inverse condition when accompanied by these domain margins. The map under consideration is $\mathcal{T}^{-1}\Psi_\Delta\mathcal{T}$, rather than discretization after a nonlinear coordinate substitution.

For a method of order p_{int}, analytic consistency and short-time perturbation estimates on smaller domains give a map defect $C_1\Delta\varepsilon + C_2\Delta^{p_{\text{int}}+1}$. Require an invertible mixed endpoint projection and a globally periodic primitive. The resulting mixed generating equations are

$$\begin{aligned} I_+ &= I_- - \partial_\phi G_\Delta(I_+, \phi_-), \\ \phi_+ &= \phi_- + \Delta\omega(I_+) \\ &\quad + \partial_I G_\Delta(I_+, \phi_-), \end{aligned} \tag{S73}$$

with $\omega = \nabla h_0$ and the domain-reduced bound

$$\|G_\Delta/\Delta\| \le C_3\varepsilon + C_4\Delta^{p_{\text{int}}}. \tag{S74}$$

Assume uniform twist bounds

$$\begin{aligned} \iota\|v\|_\infty \le \|D_I\omega(I)v\|_\infty \le \Theta\|v\|_\infty, \\ 0 < \iota \le \Theta. \end{aligned} \tag{S75}$$

For the sufficient analytic-map route of Shang (1999), choose $\tau_D > n+1$, $0 < \Delta \le 1$ and $0 < \gamma_D \le \min(1, \rho_I \Theta/2)$, where ρ_I is the complex action radius. Retained frequencies must have distance at least $2\gamma_D$ from the frequency-domain boundary and satisfy all integer discrete divisors in the main text. On the theorem's smaller complex domain require

$$C_3\varepsilon + C_4\Delta^{p_{\text{int}}} \le \delta_0 \gamma_D^2 \iota^2 \Theta^{-3}. \tag{S76}$$

Then the analytic map theorem gives the corresponding invariant tori; its domain geometry and boundary regularity assumptions are also needed for positive measure. The constants depend on dimension, exponent, widths, derivatives and chart bounds. They have not been evaluated for the current three-dimensional model. Finite solve residuals, dissipative forces and switching do not automatically satisfy this exact-map hypothesis.

2.5 KAM transfer and the inherited two-mode result

For a finite-smoothness formulation, take $n \ge 2$, $\tau > n-1$ and integer $\ell > 2(\tau+1)$. In one canonical action–angle chart, assume $h_0 \in C^{\ell+2}$ and $\nabla h_0 : B \to \Omega$ is a diffeomorphism with bounded C^ℓ inverse. Restrict to frequencies satisfying both $|k\cdot\omega| \ge \gamma |k|_1^{-\tau}$ for every nonzero integer k and $\operatorname{dist}(\omega, \partial\Omega) \ge \gamma$. With the domain, norms and threshold c_K of the applicable finite-smoothness theorem (Bounemoura, 2020), let $\beta_\theta \ge \|P_\theta - \bar{P}_\theta\|_{C^\ell}$ and suppose $\beta_\theta + E_\theta < c_K\gamma^2$. Since

$$P_* - \bar{P}_* = (P_\theta - \bar{P}_\theta) - e_\theta, \tag{S77}$$

the two perturbations satisfy the threshold separately. The theorem provides invariant Lagrangian tori for both Hamiltonians, with potentially different embeddings. Under its frequency-boundary hypotheses, the excluded-measure estimate is $C\gamma$, not $C\gamma^2$. This statement requires the chart and physical perturbation bounds in addition to learning error. The present sample loss is a squared-gradient loss; it is not a numerical evaluation of the required $p > d$ bound without further estimates.

The inherited planar full-cosine model has quartic action Hessian

$$\begin{aligned} A_{\text{old}} &= \frac{\operatorname{diag}(d_j^4) - 2vv^T}{8}, \\ v_j &= d_j^2, \\ \det A_{\text{old}} &= (1-2n)8^{-n}\prod_j d_j^4. \end{aligned} \tag{S78}$$

The matrix determinant lemma proves nondegeneracy when all $d_j \ne 0$; positivity is unnecessary. This algebra belongs to that cosine model and does not include the new variable-mass cubic correction.

The frozen two-degree-of-freedom finite-amplitude calculation additionally uses a specified Fourier embedding, a complex strip, aliasing and tail bounds, a twist inverse and an a-posteriori smallness test. The archived text reports strip defect below 2.731×10^{-77}, correction below 2.0894×10^{-41} and a tip witness above 1.8977904 mm for the short-trajectory COMSOL fit. The original Fourier/Arb/input archive is unavailable here, so these historical numbers are not independently recertified. The Python three-seed fit, full-FE integral fit and short-trajectory fit are distinct frozen Hamiltonians and do not exchange certificates or control coefficients. The extremely thin certified tube also does not locate a given physical initial condition inside it.

The tube's primitive needs a reference action. For $J = j(t) + DF(t)^{-T}I$, let $\beta = j \cdot DF\, dt$. Lagrangianity implies $d\beta = 0$, whereas its periods need not vanish. Define $a_j = (2\pi)^{-1}\oint_{\gamma_j}\beta$. Then $\beta - a\cdot dt$ has zero periods and equals dS for a periodic S, so

$$\Psi^*\lambda - (I+a)\cdot dt = dS. \tag{S79}$$

To prove this, integrate the closed zero-period form on the universal cover; zero periods allow the primitive to descend to the torus. The unshifted map remains symplectic and preserves volume. A circular torus has $\beta = r^2 dt/2$ with nonzero period πr^2, showing why unshifted exactness does not follow from closedness.

The inherited controlled 64-mode planar lift assumes positive low-mode momentum Hessian, convex momentum fibres, an invertible two-port magnetic interface, a nonvanishing geometric control factor and realizable inputs. Correct Legendre inversion makes the two controlled low modes follow the prescribed Hamiltonian. The remaining damped linear modes have Fourier responses

$$a^*_{j,k} = \frac{b_j F_k + t_j U_k}{\omega_j^2 - (k \cdot \Omega)^2 + i\alpha(k \cdot \Omega)}, \qquad \alpha > 0. \tag{S80}$$

The denominator is positive at zero frequency and has nonzero imaginary part at every nonzero real frequency. Analytic forcing thus gives an analytic response on a slightly smaller strip, and deviations satisfy exponentially stable homogeneous equations. This lifts a two-dimensional invariant torus into the controlled 128-dimensional planar state space; it does not make the whole closed loop Hamiltonian or certify a positive-volume 128-dimensional KAM family. In its separate planar S-path task, CHNN tip feedback achieved 6.419 μm error, versus 6.207 μm for the quadratic model, 4.124 μm for static physical inversion and 25.308 μm for modal output feedback. That historical task used base feed and excluded flow and contact. The negative comparison and its planar scope remain part of the evidence.

3 Evidence selection and reproducibility

The repository base is `1c73a580441200d6594e2a2a96374edc1c0e98e6`. The manuscript branch adds only the paper directory. Source comments in the article and proof files identify the existing derivations; the accompanying CSV manifests give exact numerical fields and SHA-256 identities. Scientific programs are not executed by the manuscript build. The plot builder reads accepted frozen JSON/CSV/NPZ outputs and records every plotted series.

3.1 Accepted and rejected calculations

The accepted transport results are the `inverse_verified` runs, checked against reverse differentiation and finite differences. The earlier nested automatic-differentiation path through a linear solve is not used as high-derivative evidence. The final coil-force comparison and sampled-memory diagnostics use the force-allocator warm-start correction. The final spatial-rod set uses the analytic feasibility repair. Earlier failures remain in their source directories. No old output is overwritten, relabeled as a success, or averaged into the accepted series.

The same-state coil quadrature comparison, the short feedback trajectories, the conservative integrator test, the learning labels, and the environmental rod have different purposes and, where specified, different physical branches. Each comparison is internally matched. They do not jointly constitute a single learned, constrained, Birkhoff-integrated environmental trajectory. Field samples certify only the recorded points; local polynomial bounds are not outward-rounded interval certificates.

3.2 Reproduction entry points

The paper bundle provides `README.md`, `tools/build.sh` and `tools/build_figures.py`. The scientific reproduction commands and dependencies remain in the frozen Stage 3 and Stage 5 READMEs; the paper does not claim to have rerun them. The principal manuscript manifests are:

- `evidence/theory_claims.csv`: assumptions, equations and proof sources;
- `evidence/physics_claims.csv`: physical parameters, force and rod results;
- `evidence/numerical_claims.csv`: integration, representation and solver diagnostics;
- `evidence/learning_claims.csv`: training, inversion, allocation and resonance results;

- `evidence/figure_sources.csv` and `plotted_data.json`: accepted series and source hashes;
- `evidence/reference_verification.csv`: bibliographic provenance and limits of current online verification.

Historical frozen certificates and earlier training are distinct from the new Stage 5 checkpoints. Where the original historical weights, Fourier assets or complete data archive are absent from the lightweight bundle, their absence prevents exact reproduction and is retained as such. The existing Stage 5 results provide their own configurations and outputs; they are not replacements for missing historical files.

References

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